

Headline XIV: the per-stratum posterior expectation in general dimension

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Abstract

Unit 205 takes the quotient the paper actually cares about. With numerator amplitude φc (observable times density) and denominator amplitude $c > 0$, both dressed by the same phase ξ , Headline XIII gives both leading terms at the same scale, and the denominator coefficient is strictly positive, so

$$\frac{\int_{(0,1]^d} \varphi(u) c(u) u^h e^{-\beta N^2 u^{2k} + \beta N u^k \xi(u)} du}{\int_{(0,1]^d} c(u) u^h e^{-\beta N^2 u^{2k} + \beta N u^k \xi(u)} du} \longrightarrow \frac{\int \varphi(\pi u) c(\pi u) J_p(\xi(\pi u)) w(u) du}{\int c(\pi u) J_p(\xi(\pi u)) w(u) du}$$

with $w(u) = \prod_{i \notin J} u_i^{h_i - p k_i}$ (headline_phase_posterior): the per-stratum posterior expectation of φ concentrates on the face $u_J = 0$ with density $c J_p(\xi) w$. In the equal-ratio case the face is the corner and the limit is $\varphi(0)$ (headline_phase_posterior_equal). Zero sorry/axiom; 9082 jobs.

1 The things formalised

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theorem faceProj_mem_closedCube (hu : u  closedCube d) : faceProj h k l u
  closedCube d
theorem continuous_faceProj : Continuous (faceProj h k l)
theorem phaseMoment_pos (h) (hp : 0 < p) : 0 < phaseMoment  p a
theorem integral_residualWeight_pos (hk) (hl) (h) (hmin) :
  0 <  u in unitBox d, residualWeight h k l u
theorem faceNorm_pos (hk) : 0 < faceNorm h k l
theorem phaseCoeff_pos (hk) (hl) (h) (hmin) (h) (hc) (hcpos :  x  closedCube d,
  0 < c x) :
  0 < phaseCoeff h k l  c
theorem phase_posterior_tendsto ... (hcpos) :
  Tendsto (fun N => ( ( x * c x) * ( x^h * quadKernel  ( x) (N  x^k)))
    / c x * ( x^h * quadKernel  ( x) (N  x^k))) atTop
    ( (phaseCoeff h k l  (fun x => x * c x) / phaseCoeff h k l  c))
theorem headline_phase_posterior ... : ... atTop
  ( (( u in unitBox (m+1), ( ( u) * c ( u) * phaseMoment  (2l) ( ( u)))
    * w u)
    / u in unitBox (m+1), (c ( u) * phaseMoment  (2l) ( ( u))) * w u))
theorem phaseCoeff_equal (hratio) :
  phaseCoeff h k l  = 0 * phaseMoment  (2l) ( 0) / (m! * i, k i)
theorem headline_phase_posterior_equal (hratio) (hcpos) : ... atTop ( ( 0))

```

2 Mathematics

Positivity of the denominator coefficient: the face density $G(u) = c(\pi u)J_p(\xi(\pi u))$ is continuous on the closed cube (`faceProj` is continuous: each coordinate is either 0 or a coordinate projection) and positive there ($c > 0$, $J_p > 0$ as an integral of a positive integrand over $(0, \infty)$), so it has a positive minimum $G(x_0)$; then $\int Gw \geq G(x_0) \int w > 0$, where $\int_{(0,1]^d} w > 0$ follows from the zero-phase amplitude theorem at constant amplitude (`faceLeadConst` · $\int w = \text{monomialMixedConst} > 0$). The quotient theorem is then `Tendsto.div` of the two Headline XIII limits, with the common scale $N^{-p}(\log N)^{m-1}$ cancelling exactly for $N > 1$ (`div_div_div_cancel_right`). The prefactor $2^{m-1} \cdot 2 \cdot \text{faceNorm}$ cancels in the quotient, leaving the ratio of face integrals; in the equal-ratio case the face is the corner and $\varphi(0)c(0)J_p(\xi(0))/(c(0)J_p(\xi(0))) = \varphi(0)$.

This is the general- d , deterministic-phase, single-chart form of the paper's statement that posterior expectations localise to the exceptional divisor: the observable is read on the face $u_J = 0$, weighted by the density c , the dressed normal moment $J_p(\xi)$ of the phase, and the residual weights of the non-minimising coordinates. It matches the $d = 2$ chart posterior limit $Z[\varphi]/Z[1] \Rightarrow y_{\varphi,00}/y_{1,00}$ of unit 159 (there the amplitudes were independent Taylor arrays; here $y_{\varphi,00} = \varphi(0)c(0)$, $y_{1,00} = c(0)$).

3 Lean notes

- `faceProj_mapsTo` only maps the half-open box into the closed cube; the closed-cube version needed here is a two-line lemma (`faceProj_mem_closedCube`).
- The quotient identity $(a/K)/(b/K) = a/b$ then $(\varphi cJ)/(cJ) = \varphi$: rather than `field_simp` (which left a goal), the three rewrites `div_div_div_cancel_right`, `mul_div_mul_right`, `mul_div_cancel_right` close it exactly.
- `IsCompact.exists_isMinOn` needs a nonemptiness witness; $0 \in [0,1]^d$ is 0, `fun i _ => le_refl, zero_le_one`.
- An unused hypothesis (`hk` in `phaseCoeff_equal`) is flagged by the linter; drop it from the statement.

4 Status

Headlines XIII–XIV close the deterministic general- d story at leading order: dressed integral, cutoff, and posterior quotient. Remaining gaps (report v2 §7): random phases in general dimension, subleading terms, chart assembly.